English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

The maximum total for all homework assignments during the semester is 50 points, corresponding to a bonus of 0.5 towards the exercise-class grade.

Problem 1. (3 points) Let X, Y be random numbers from $\{1, \ldots, n\}$[1], and let p_n be the probability that $X + Y$ is a perfect square. Find

$$\lim_{n\to\infty} \sqrt{n} p_n.$$

Problem 2. ($6 \times 2 = 12$ points)

1. One card is drawn from each of 10 standard decks of 52 cards. Find the probability that the resulting 10-card hand:
   (a) contains no repeated cards;
   (b) contains at least three aces;
   (c) contains four clubs, three diamonds, two hearts and one spade;
   (d) contains exactly 4 more black cards than red cards, given that there are more black than red cards.
2. All items in one batch are functional; in another batch, 1/4 are defective. An item from a randomly selected batch turns out to be functional. Calculate the probability that a second randomly chosen item from the same batch is defective, if the first item was returned to its batch after inspection.
3. What is the probability that the roots of $x^2 + ax + b = 0$, $a, b \in [0, 1]$, are real?
4. Two faces each of a fair die are coloured white, green and red. Roll it twice. Let X count white outcomes and Y count red outcomes. Find their joint distribution, determine whether they are independent, and find their covariance, $\mathbb{P}(X = 1 \mid Y = 1)$ and $\mathbb{P}(X > Y)$.
5. Let X take values in $\mathbb{N}_0$, and let
$$g_X(s) := \mathbb{E}s^X = \sum_{k=0}^{\infty} s^k \mathbb{P}(X = k)$$
be its generating function.
   (a) Let $Y := 3X$ and $Z = X_1 + X_2$, where $X_1, X_2 \sim X$ are independent. Express g_Y and g_Z in terms of g_X.
   (b) Let $X \sim Ge(p)$, meaning $\mathbb{P}(X = k) = p(1-p)^k$ for $k \in \mathbb{N}_0$. Calculate g_X and use it to find $\mathbb{E}(X)$ and $D(X)$.
6. Let X, Y be independent random variables with
$$\mathbb{P}(X = -1) = \mathbb{P}(X = 1) = \mathbb{P}(Y = 1) = 2\mathbb{P}(Y = 3) = 2\mathbb{P}(Y = 5) = \frac{1}{2}.$$
Let $Z_1 := 2X + Y + 1$, $Z_2 := XY$, and $Z_3 = X^Y$. Find their expectations and variances.

Problem 3 (5 points)**.** Let $n \geq 3$ and p be natural numbers. Consider the following game with players $A_1, \ldots, A_n$:

- Each player writes predictions for $(2p + 1)$ tosses of a standard coin (probability 1/2 for each of heads and tails), using H for heads and T for tails.
- The coin is tossed $(2p + 1)$ times, and the players with the most correct predictions split a prize of $n!$ leva.

[1] That is, independent and uniformly distributed on this set.

Example: $n = 4$, $p = 1$, predictions HTH, HHH, TTT, HHT. With outcome TTH, players 1 and 3 each predict 2 correctly and split 4!, receiving 12 leva each. For $n = 4$, $p = 1$, the same predictions and outcome TTT, player 3 predicts all 3 correctly and wins $4! = 24$ leva. For $i \in [n]$, define

- $X_i =$ "the number of outcomes correctly predicted by A_i";
- $G_i =$ "player A_i's winnings";
- $q_k = \mathbb{P}(X_i = k)$, $r_k = \mathbb{P}(X_i \leq k)$.

Our aim is to calculate player A_1's expected winnings, $\mathbb{E}(G_1)$, under two strategies.

Part 1. Preliminary considerations.

1. * As usual, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. What set would you propose for Ω?
2. What is the distribution of X_i?
3. Calculate r_p.
4. Recall that $X(\Omega)$ denotes all images of elements of Ω under X. For example, $X_1(\Omega) = \{0, 1, \ldots, 2p+1\}$, since A_1 can predict between 0 and $2p+1$ outcomes correctly. If $\mathbb{P}(A) > 0$, define conditional expectation by

$$\mathbb{E}(X \mid A) = \sum_{k \in X(\Omega)} k\mathbb{P}(X = k \mid A).$$

   Prove that if $B_1, \ldots, B_s$ form a partition into events of nonzero probability, then

$$\mathbb{E}X = \sum_{k=1}^{s} \mathbb{E}(X \mid B_k)\mathbb{P}(B_k).$$

   (Expand $\mathbb{E}X$ by definition, apply the law of total probability, and interchange the sums.)

Part 2. All players play randomly. In this part, assume that all players choose heads or tails randomly with probability 1/2, independently of each other.

1. What is $G_1(\Omega)$, the set of possible winnings for A_1?
2. For every $k \in X_1(\Omega)$ and $j \in [n]$, prove that

$$\mathbb{P}\left(G_1 = \frac{n!}{j} \,\middle|\, X_1 = k\right) = \binom{n-1}{j-1}(r_k - r_{k-1})^{j-1} r_{k-1}^{n-j}.$$

3. Prove that, for $k > 0$,

$$\mathbb{E}(G_1 \mid X_1 = k) = \frac{(n-1)!(r_k^n - r_{k-1}^n)}{q_k}.$$

   Note that the formula also holds for $k = 0$ if we define $r_{-1} = 0$. *(Expand using the preceding part and use the binomial theorem.)*
4. Conclude that $\mathbb{E}(G_1) = (n-1)!$. *(Consider the partition $\{X_1 = k\}$, $k \in \{0, 1, \ldots, 2p+1\}$.)*
5. Is the conclusion of the preceding part expected? We could also have obtained it differently: can we calculate $G_1 + \cdots + G_n$? Use linearity of expectation and symmetry to obtain $\mathbb{E}(G_1)$ again.

Part 3. A_1 and A_2 cooperate; all other players play randomly. Assume that everyone except A_2 plays randomly. A_2 knows A_1 and predicts the opposite of A_1; for example, HHT for A_1 means TTH for A_2. If either wins, they share the winnings equally. Write

- G', G_1', G_2' for their combined and individual winnings, respectively, so $G' = G_1' + G_2'$;
- $Y =$ "the larger number of correct predictions by A_1 and A_2" (for predictions HHT, TTH and outcome HHH, $Y = 2$);

- as before, for $k \in \{1, 3, 4, \ldots, n\}$, $q_k = \mathbb{P}(X_i = k)$ and $r_k = \mathbb{P}(X_i \leq k)$.

1. Prove that $Y(\Omega) = \{p+1, \ldots, 2p+1\}$.
2. Prove that for $k \in \{p+1, \ldots, 2p+1\}$, $\mathbb{P}(Y = k) = 2q_k$.
3. Prove that for $k \in \{p+1, \ldots, 2p+1\}$ and $j \in \{0, \ldots, n-2\}$,

$$\mathbb{P}\left(G' = \frac{n!}{j+1} \,\middle|\, Y = k\right) = \binom{n-2}{j}(r_k - r_{k-1})^j r_{k-1}^{n-2-j},$$

   and consequently, using the binomial theorem,

$$\mathbb{E}(G' \mid Y = k) = n(n-2)!\frac{r_k^{n-1} - r_{k-1}^{n-1}}{q_k}.$$

4. Conclude that

$$\mathbb{E}(G') = 2n(n-2)!\left(1 - \frac{1}{2^{n-1}}\right),$$

   and check that this strategy is better for A_1, A_2 than the strategy in Part 2.
5. Calculate $\mathbb{E}(G_i)$ for $i \in \{3, \ldots, n\}$.